NATIONAL RESEARCH UNIVERSITY
HIGHER SCHOOL OF ECONOMICS

Faculty of Computer Science
Bachelor's Programme "HSE and University of London Double Degree Programme in Data Science and Business Analytics"

Research project report

on the topic Numerical Algorithms of Linear Algebra

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Abstract

The main focus of the programming project is to study matrix decomposition algorithms from a theoretical point of view with its subsequent implementation on the C++ language.

1 Introduction

Linear Algebra provides extremely useful tools for computer science and other engineering fields. However, even the simplest algorithms of linear algebra cannot be directly used in practice due to inefficiencies. Consider the matrix multiplication algorithm: in its naive implementation the complexity is $O(n^3)$, so in order to multiple two 1000x1000 matrices, you would need a billion operations!

In order to create faster algorithms a field of Numerical Analysis has been developed recently. As one of its achievements I can mention Coppersmith Winograd matrix multiplication algorithm with complexity $O(n^{2.376})$[1], which reduced the previous number of operations to just 13mln.

The goal of my programming project is to learn modern algorithms of Numerical Analysis with a focus on matrix decomposition algorithms, which represent a matrix in a certain way for further computation. In order to have a profound understanding of each of the algorithms, it is necessary to implement a C++ library from scratch.

The library itself will contain a class for matrix representation with implementation of matrix arithmetic and other necessary methods. Decomposition algorithms will be implemented based on the aforementioned matrix class. The whole library will be tested using Googletest.

The information for the theoretical part of the project is taken from the books, which are cited at the end of the paper. Citation of exact chapters is used throughout the theoretical part when needed.

The result of the project will be a C++ library LinAlgTools with the following functionality:

- QR Decomposition
- SVD Decomposition
- Spectral Decomposition

Theoretical part can help you understand the underlying theory of each of the algorithms. The technical part of the project is described in detail in the Library LinAlgTools section of the paper. The results of the testing using Googletest are documented in Testing section.

[1] In general the algorithm is not widely used due to its limitations on matrix conditions

2 Literature review

3 Description of functional and non-functional requirements for the program project

4 Theorerical part

5 Library LinAlgTools

6 Testing

7 Conclusion

References